Fantasy Baseball Draft Optimization

OR114-2 Final Project, Group 4

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1 Introduction

Fantasy baseball drafts require managers to build a competitive roster before the season even begins. In a snake draft, each manager owns only a fixed sequence of picks, while the available player pool is continually reshaped by opponents' selections and by market expectations summarized through average draft position (ADP). The practical decision is therefore not simply which players are strongest in isolation, but which players should be selected now before positional needs and market timing make better options disappear.

This decision is challenging because draft value is interdependent. A high-projection player may not be the best choice if he occupies an already abundant position, while a slightly lower-projection player may be more valuable if he protects a scarce roster slot. In practice, fantasy managers often rely on ranking lists or manual "best-player-available" judgments. However, these shortcuts do not explicitly account for how one pick changes the value of the remaining roster. The decision maker faces a complicated environment: opponents draft players, ADP creates market timing constraints, players have multi-position eligibility, and strict roster rules must be satisfied.

We study this problem as an operations research decision-support task. The core trade-off is between projected points and positional scarcity, as well as exact solution quality versus runtime. An ADP-aware integer linear program (ILP) provides an exact benchmark for smaller instances, while a Direct Greedy approach serves as a simple baseline. Our proposed method, *Opportunity Cost Greedy*, estimates the cost of waiting by comparing the current value of available players with the value likely to remain at the next owned pick. The goal is to preserve the strategic logic of the exact model while remaining computationally fast enough for larger draft settings.

The report evaluates these methods on controlled synthetic instances and real fantasy baseball data. The main message highlights a scalability trade-off: exact optimization gives a strong benchmark but becomes computationally expensive as draft instances grow, whereas Opportunity Cost Greedy remains practical and consistently improves on the direct greedy baseline.

2 Problem Description

We consider a fantasy baseball snake draft in which a manager owns a fixed set of picks determined by draft order and the number of teams. At each owned pick, the manager must choose one available player from the remaining pool. Each player has a projected fantasy value, one or more eligible roster positions, and an ADP estimate that reflects when that player is expected to be drafted in the market.

The manager's goal is to construct a final roster that satisfies all positional requirements while maximizing total projected points. A player can only be drafted if he is still available at the current pick, and a drafted player can only be assigned to an eligible roster slot. The roster is not just a list of best individual names; it is a feasible lineup structure that must include the required number of specific positions.

The base roster used in the experiments consists of 16 slots. The utility slot (`Util`) is hitter-only, meaning pitchers cannot be assigned to it. The same roster rules are used by the ILP, Direct Greedy, and Opportunity Cost Greedy, keeping the comparison fair.

Table 1: Base fantasy baseball roster requirements.

Position	C	1B	2B	3B	SS	OF	Util	SP	RP
Slots	1	1	1	1	1	3	1	5	2

This problem is sequential in appearance but integrated in structure. A choice made early in the draft affects what remains possible later. The core trade-off is between immediate value and future flexibility: taking the best current player may ignore the opportunity cost of leaving a critical position unresolved, while drafting for scarcity too early may sacrifice overall points. By explicitly defining market availability through ADP and treating draft planning as an integrated roster-construction problem, we transform a subjective drafting strategy into a rigorous optimization problem.

3 Model Formulation

Let I be the set of players, P the set of roster positions, and K the set of owned draft picks. Each player $i \in I$ has projected points v_i, ADP adp_i, and a set of eligible positions $E_i \subseteq P$. Each roster position $p \in P$ has a required count r_p. The draft order determines the pick number associated with each owned pick $k \in K$, denoted as S_k.

We define the following decision variables:

- $y_i \in \{0,1\}$: equal to 1 if player i is drafted, 0 otherwise;
- $x_{ip} \in \{0,1\}$: equal to 1 if player i is assigned to position p, 0 otherwise;
- $z_{ik} \in \{0,1\}$: equal to 1 if player i is selected with owned pick k, 0 otherwise.

The objective is to maximize total projected roster points:

$$\max \sum_{i \in I} v_i y_i$$

The model includes the following constraints. First, each owned pick must select exactly one player:

$$\sum_{i \in I} z_{ik} = 1, \quad \forall k \in K$$

Second, each player can be selected at most once across all picks:

$$\sum_{k \in K} z_{ik} \leq 1, \quad \forall i \in I$$

Third, drafting a player implies that he is assigned to exactly one roster position:

$$\sum_{p \in P} x_{ip} = y_i, \quad \forall i \in I$$

Fourth, each drafted player may only be assigned to an eligible position:

$$x_{ip} \leq \mathbf{1}(p \in E_i), \quad \forall i \in I, p \in P$$

Fifth, each roster position must be filled exactly to its requirement:

$$\sum_{i \in I} x_{ip} = r_p, \quad \forall p \in P$$

Finally, to enforce draft availability, we use an ADP-based restriction. A player cannot be chosen at pick k if the current overall pick S_k occurs too late relative to his expected market position plus a tolerance buffer δ:

$$z_{ik} = 0 \quad \text{if } S_k > \text{adp}_i + \delta, \quad \forall i \in I, k \in K$$

All decision variables y_i, x_{ip}, z_{ik} are strictly binary. The ADP buffer parameter δ controls how permissive the draft market is. This exact formulation is highly effective but becomes computationally expensive as the instance size (number of players and roster slots) grows.

4 Algorithms

To address the scalability limits of the exact formulation, this project compares the exact benchmark against two heuristic methods.

4.1 ADP-aware ILP (Exact Benchmark)

The ADP-aware ILP described in Section 3 is solved using Gurobi. It acts as the exact benchmark for optimality-gap calculations on solvable instances. Its primary strength is absolute solution quality, acting as a planning oracle. However, its weakness is scalability: as the player pool and roster requirements grow, the number of binary variables increases rapidly, leading to prohibitive runtimes.

4.2 Direct Greedy (Baseline)

Direct Greedy is the simplest baseline heuristic. At each owned pick, it looks at all open roster positions, identifies the scarcest position, and selects the best currently available player for that role. It is fast and interpretable but myopic: it makes local "best-player" choices without explicitly reasoning about the opportunity cost of waiting or future availability.

4.3 Opportunity Cost Greedy (Proposed Method)

Opportunity Cost Greedy is our proposed strategic heuristic. At each owned pick, the algorithm:

1. Expires players whose ADP availability window has already passed.
2. Identifies open roster positions.
3. Evaluates the best *currently available* player for each open position.
4. Estimates the best player *likely to remain available* at the manager's *next* owned pick.
5. Computes the **opportunity cost** (delay-cost calculation) as: Current Value − Expected Future Value.
6. Employs a fallback logic to force priority on positions where current or future supply is critically low.
7. Drafts the feasible player-position pair with the highest opportunity-cost urgency score.

This method captures the strategic logic of the ILP—balancing immediate value with future positional scarcity—but executes sequentially, allowing it to scale gracefully to large instances.

5 Data Collection and Generation

The project uses real fantasy baseball inputs for practical validation and synthetic instances for controlled evaluation. The real-world instances use 2026 player projections and ADPs sourced from Yahoo and FanGraphs, confirming that the framework functions smoothly on realistic data.

However, the main experimental evidence comes from synthetic instances. Synthetic data allows us to control and isolate operational factors that affect draft difficulty: player point distributions, position flexibility, ADP market noise, and scale. Points are generated as objective coefficients v_i, positions as eligibility sets E_i, and ADP as the market-availability signal. ADP is generated by ranking players by projected points and adding Gaussian noise (σ_{adp}).

To systematically evaluate our algorithms, we designed 6 experiment families (N1 to N6) driven by these operational factors. The parameter configurations for all scenarios are summarized in Table 2.

Table 2: Summary of the 6 synthetic experiment scenarios and their specific parameter configurations. Variables in bold represent the varied factors in each specific scenario block.

Scenario	Main Focus	Points Dist.	Positions Dist.	Scale & Demand	ADP (σ_{adp}, δ)	Reps
N1	Baseline	normal	roster-ratio	Scale=1, D=3	(30, 0)	10 seeds
N2	Points Dist.	**varied (3 levels)**	roster-ratio	Scale=1, D=3	(30, 0)	10 seeds
N3	Positions Dist.	normal	**varied (4 levels)**	Scale=1, D=3	(30, 0)	10 seeds
N4	Scaling	normal	roster-ratio	**S=1-3, D=1-10**	(30, 0)	10 seeds
N5	ADP Noise	normal	roster-ratio	Scale=1, D=3	**varied σ, $\pm\delta$**	10 seeds
N6	Stress Test	high-low	single-position	**Large Scale**	high noise	1 run

Detailed Factor Meanings:

- **Points Distribution (N2)**: *normal* clusters values around the middle; *uniform* spreads values evenly; *high-low* concentrates value into a small elite group, making misses extremely costly.
- **Position Distribution (N3)**: *point-flexible* rewards strong players with multi-position eligibility; *single-position* removes flexibility entirely, making early greedy mistakes hard to repair; *roster-ratio* aligns position supply with roster demand.
- **Scaling (N4 & N6)**: *roster scale* multiplies the 16-slot base requirement. *Player-demand ratio (D)* adjusts the pool size relative to league demand (e.g., D=1 means almost every player is drafted, creating extreme scarcity).
- **ADP Noise (N5)**: $\sigma_{\text{adp}} = 0$ means ADP perfectly follows value. Larger σ_{adp} makes the market unpredictable, creating traps and bargains.

6 Performance Evaluation

We evaluate the algorithms using the **optimal gap ratio** relative to the ADP-aware ILP benchmark:

$$\text{gap ratio}_m = \frac{Z_{\text{ILP}} - Z_m}{Z_{\text{ILP}}} \times 100\%$$

where Z_{ILP} is the objective value of the exact model, and Z_m is the heuristic's objective value. Smaller gaps indicate better performance. For N6, we also measure runtime (seconds) against approximate IP variable counts to prove scalability.

6.1 N1 to N3: Baseline, Points, and Positions

Table 3 summarizes the results for scenarios N1 through N3. In the **N1 Baseline**, Opportunity Cost Greedy (OCG) achieves a gap of 1.92%, a clear improvement over Direct Greedy's 4.79%.

In **N2**, the *high-low* point distribution creates a highly star-sensitive environment. Direct Greedy's gap spikes to 8.39% because it myopically misses elite players, while OCG maintains a much tighter 2.87% gap by explicitly pricing the delay cost of missing stars.

In **N3**, the *single-position* scenario poses the strictest test for flexibility. Even when early mistakes are hardest to repair, OCG outperforms Direct Greedy significantly (0.94% vs 5.11%).

Table 3: N1, N2, and N3 results. Values are optimal gap ratio (% $\pm$ std).

Scenario	Level	Direct Greedy	Opportunity Cost Greedy
N1: Baseline	normal	4.79 ± 1.18	1.92 ± 0.57
N2: Points Dist.	uniform	3.24 ± 0.77	1.44 ± 0.59
N2: Points Dist.	high-low	8.39 ± 2.40	2.87 ± 1.32
N3: Pos Dist.	uniform-by-type	4.89 ± 0.62	1.07 ± 0.43
N3: Pos Dist.	point-flexible	5.36 ± 1.66	1.34 ± 0.68
N3: Pos Dist.	single-position	5.11 ± 1.48	0.94 ± 0.65

6.2 N4 and N5: Scaling and ADP Uncertainty

In **N4**, we scale the problem size. As seen in Table 4, OCG is consistently superior across all combinations. When supply is tightest (Demand=1), Direct Greedy's gap reaches 8.13%, whereas OCG restricts the loss to 3.02%.

In **N5**, as ADP noise (σ_{adp}) increases, the draft market becomes unpredictable. While both heuristics degrade under high noise, OCG remains substantially better because it utilizes the next-pick availability probability rather than treating all open positions equally.

Table 4: N4 (Scaling) and N5 (ADP Uncertainty) results. Values are optimal gap ratio (% $\pm$ std).

Scenario	Level	Direct Greedy	Opportunity Cost Greedy
N4: Scaling	Scale 1, Demand 1	8.13 ± 1.64	3.02 ± 0.95
N4: Scaling	Scale 1, Demand 10	3.39 ± 0.96	1.23 ± 0.38
N4: Scaling	Scale 2, Demand 1	6.65 ± 0.40	2.46 ± 0.63
N4: Scaling	Scale 2, Demand 10	2.34 ± 0.53	0.81 ± 0.40
N4: Scaling	Scale 3, Demand 1	4.82 ± 0.40	2.69 ± 0.55
N4: Scaling	Scale 3, Demand 10	1.88 ± 0.22	0.48 ± 0.17
N5: ADP Noise	$\sigma_{\text{adp}} = 0$	1.72 ± 0.70	0.42 ± 0.17
N5: ADP Noise	$\sigma_{\text{adp}} = 10$	3.34 ± 0.91	0.99 ± 0.45
N5: ADP Noise	$\sigma_{\text{adp}} = 30$	5.03 ± 1.28	1.80 ± 0.75
N5: ADP Noise	$\sigma_{\text{adp}} = 60$	5.73 ± 1.35	2.43 ± 1.38
N5: ADP Noise	$\sigma_{\text{adp}} = 100$	4.50 ± 1.48	2.15 ± 1.16

6.3 N6: Large-Scale Stress Test

N6 is designed to evaluate scalability by pushing the models into regimes where exact optimization becomes impractical. To make the stress realistic, N6 uses the *high-low* points and *single-position* settings. Table 5 reports a single representative run per stress level.

Table 5: N6 Large-Scale Stress Test Results. Run times are in seconds. The Approximate Variable count demonstrates the exponential growth of the ILP search space.

Instance	Approx. Variables	Players	Roster	DG Time	OCG Time	ILP Time / Status
stress_small	334,080	5,760	48	0.790	0.852	4.235 / OPTIMAL
stress_medium	710,400	9,600	64	0.951	1.144	12.230 / OPTIMAL
stress_large	2,289,600	21,600	96	1.981	2.344	54.267 / OPTIMAL
stress_xlarge	10,880,000	64,000	160	6.172	10.360	276.895 / OPTIMAL
timeout_target	49,029,120	184,320	256	17.760	23.286	1869.805 / TIMEOUT

As the approximate variable count grows to nearly 50 million, the ILP runtime scales exponentially, eventually hitting the 1800-second wall-time limit in `timeout_target`. In stark contrast, Opportunity Cost Greedy finishes the exact same massive instance in just 23.286 seconds. This confirms that our heuristic is not just an approximation, but the only practical operational tool at massive scales.

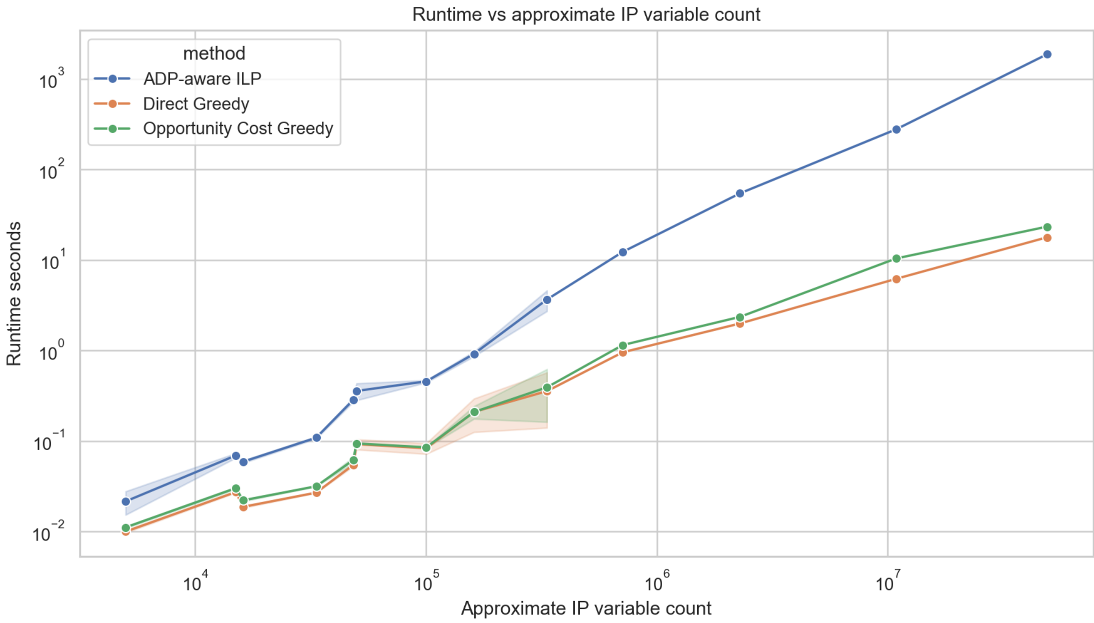

Figure 1: Runtime (seconds) versus approximate IP variable count for N6.

6.4 Real-Data Validation

To ensure practical applicability, we validated our algorithms on 2026 Yahoo and FanGraphs datasets (252 draft cases each). As shown in Table 6, the ranking of the methods matches the synthetic findings perfectly. Opportunity Cost Greedy substantially reduces the optimality gap left by the naive Direct Greedy baseline on realistic draft settings.

Table 6: Real-data validation summary (Gap measured in absolute projected points).

Dataset	Method	Mean Objective	Mean Gap	Max Gap
Yahoo 2026	ADP-aware ILP	7917.209	0.000	0.000
Yahoo 2026	Opportunity Cost Greedy	7767.732	149.477	255.700
Yahoo 2026	Direct Greedy	7662.866	254.343	373.000
FanGraphs 2026	ADP-aware ILP	14642.927	0.000	0.000
FanGraphs 2026	Opportunity Cost Greedy	14320.135	322.792	508.000
FanGraphs 2026	Direct Greedy	14097.337	545.590	644.500

7 Conclusions

This project successfully models fantasy baseball snake-draft planning as an operations research problem. The primary challenge was balancing positional constraints, market timing, and draft scarcity against the computational limits of exact optimization.

Our empirical findings convey a clear message: while the ADP-aware ILP provides an undeniable exact benchmark, its exponential runtime growth renders it impractical for large-scale or real-time drafting applications. Our proposed method, **Opportunity Cost Greedy**, effectively solves this by estimating the delay-cost of waiting for future turns. Across all controlled synthetic environments—including elite-heavy distributions, rigid positional constraints, tight supply ratios, and highly noisy ADP markets—OCG consistently narrowed the optimality gap compared to the myopic Direct Greedy baseline. Furthermore, it proved its scalability by solving massive 50-million variable instances in seconds.

Limitations and Future Work: Currently, ADP serves only as a proxy for opponent behavior. In reality, human opponents do not follow ADP mechanically, and player projections carry inherent variance (e.g., injury risks). Future extensions could incorporate probabilistic opponent modeling, bench-value management, and hybrid methods that use OCG to generate an initial incumbent solution, followed by localized ILP repairs.

Overall, this project shows that while the ADP-aware ILP provides the exact benchmark for draft optimization, Opportunity Cost Greedy delivers a highly strategic, practical alternative that is fast enough to act as a real-time decision-support tool.